

August 22, 2026

Prof. Jin Xuan and Prof. Jinfeng Liu
Editors-in-Chief
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Dear Professors Xuan and Liu,

Please consider my original research article, “Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate,” for publication in *Digital Chemical Engineering*.

The manuscript develops a closed-loop design-and-operation framework for a prescribed N -stage activated sludge system connected to a secondary Clarifier. The number of stages is a fixed topology parameter, and the case study uses five continuous stirred-tank reactors. The decisions include nominal hydraulic retention time, zonal aeration, mixed-liquor recycle, return activated sludge, and waste activated sludge. One dimensionless second-order surrogate jointly predicts the mixer, every reactor outlet, Clarifier outlet component mass flows, and the ten-layer Clarifier solids profile, giving 170 response coordinates in the case study.

The five process controls and 110 reactor-and-Clarifier states are optimized together in one sparse nonlinear program. The complete 170-coordinate response is reconstructed from those physical states. All reactor and Clarifier steady-state balances, non-negativity, two positive-solids guards, and five engineering restrictions are imposed directly. An independently calibrated fidelity region and a development-leverage limit keep the reconstructed state tied to the statistical response and its training support. The engineering criterion is the sole objective.

The study builds on the component-space and invariant ideas in Selerio Jr. (2026), *An interpretable statistical surrogate of activated sludge systems that preserves mass conservation and component non-negativity*, *Digital Chemical Engineering*, 20, 100329. The estimator uses 351 dimensionless, uniquely serialized quadratic features in one full-column-rank standardized multiresponse ordinary least-squares problem. Column-pivoted QR computes the fixed coefficient matrix, and an independent singular-value decomposition verifies rank, conditioning, and coefficient agreement.

Training and validation use a coupled ASM2d-TSN-multilayer-Clarifier model. Independent development, calibration, and untouched assessment blocks contain 70%, 10%, and 20% of the 20,000 mechanistic states. The development-only estimator is never refitted after calibration or assessment. Nine deterministic CasADi-IPOPT starts solve the combined physics-constrained statistical problem. One independent exact nonsmoothed dynamic replay tests the selected decision for stability, engineering feasibility, statistical fidelity, and mechanistic branch agreement. The study therefore reports one reproducible multistart local candidate per case without claiming global optimality.

The work is aligned with the journal’s scope in digital process engineering, mathematical programming, physically constrained surrogate modeling, and computational wastewater-process operation. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I have approved its submission and declare no competing interests. Thank you for your

consideration.

Sincerely,

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Development for Technology
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